

module-05-dot-addition - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 5 — Bindu Paddhati — Dot Addition (Darśana)

Sanskrit: *Bindu Paddhati* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

A 20th-century mental-arithmetic technique for column addition.

Darśana note. This workshop condenses Module 5's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Project the problem: $47 + 38 + 26 + 91 + 53 = ?$ Time the class. Most: ~30-45 seconds. “*There’s a faster method, documented in 20th-century Indian textbooks. It’s called Bindu Paddhati. Today we learn it AND examine its honest historical claim.*”

Core idea · 15 min

Bindu Paddhati — modern reconstruction.

The method as taught is well-documented in 20th-century Indian mental-arithmetic primers (e.g., Bharati Krishna Tirthaji’s *Vedic Mathematics*, 1965; later popularizations).

Honest historical framing: - The DOT NOTATION as such is a modern (20th-c.) pedagogical convention. - The PLACE-VALUE ARITHMETIC behind it is much older (Brahmi numerals, ~3rd c. BCE; place value formalised ~Āryabhaṭa, 499 CE). - The CLAIM that this is “Vedic” is contested (S.G. Dani, *Frontline* 1993). It’s a 20th-c. arithmetic synthesis that uses Sanskrit names.

Method itself: sound. **Historical claim:** modern, not ancient.

Both can be true. Use the method; teach the honest history.

Demonstration · 10 min

Source comparison. Project two passages: (a) Tirthaji 1965 description of *Bindu Paddhati*, (b) a Sanskrit excerpt claimed to be the “original” but actually a 20th-c. Sanskrit composition. Students compare: where does the documentary evidence lie? What’s the honest dating?

Source moment · 5 min

Quoted reference. Tirthaji, *Vedic Mathematics* (1965, posthumous), in the chapter on column addition:

“The dot above the digit signifies ten. The student need not carry mentally; the dot does the carrying.”

Real source, real date. **Note:** Tirthaji’s book itself is contested as “Vedic.” Use the technique honestly: it’s a 20th-c. mental-arithmetic system, not a Vedic-period text.

Hands-on activity · 15 min

Critical historical-mathematics essay. Each student writes 200 words: “*The Bindu Paddhati is empirically useful but its claimed antiquity is unprovable. How should educators present such methods honestly? What is lost by overclaiming antiquity? What is lost by dismissing the method as not-really-ancient?*”

Reflection + take-home · 10 min

“*This module taught you a technique AND a historical critique of its claimed origin. How will you teach this honestly to a younger student?*”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — Why mental arithmetic? The setup
- **Day 02** — Single-digit + single-digit with the dot
- **Day 03** — Two-digit column addition
- **Day 04** — Three- and four-digit columns
- **Day 05** — Speed practice — Bindu Paddhati vs traditional
- **Day 06** — Why it works — place value + carry
- **Day 07** — Honest history — Tirthaji 1965, Dani 1993 critique
- **Day 08** — Bridge to computer arithmetic — carry-save addition

- **Day 09** — Mixed problem set
- **Day 10** — Project — design your own mental-math poster

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.